

COMPUTER SCIENCE TRIPOS Part IB – 2023 – Paper 5

8 Introduction to Computer Architecture (swm11)

Technology
trends and design
challenges

- (a) What is the *power wall* and why does it lead to the idea of *dark silicon*? [3 marks]

Answer: The *power wall* refers to the maximum power that a chip can draw before thermal limits (or power-supply limits) are reached. Moore's law (transistor density) has resulted in chips with billions on transistors and yet comparatively few of them can be switched due to the power wall. This leads to the idea of *dark silicon* since many of the transistors must be inactive or dark.

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- (b) What is the difference between race-to-dark (also known as race-to-sleep or race-to-halt) and dynamic voltage/frequency scaling in order to save power? [3 marks]

Answer: Race-to-dark is the idea of completing a task quickly and then going into a low power state in order to save power. This saves power by minimising the on time. In contrast, voltage/frequency scaling exploits the fact that power consumption is proportional to the square of *voltage* ($voltage^2$) whereas performance is proportional to *voltage* (so long as the transistors are not operating subthreshold). So reducing the supply voltage saves a lot of power while only reducing performance a little. So, where possible we run systems in a power efficient state and only raise the voltage when additional performance is required, e.g. where we have one sequential application to run and we cannot make use of other processor cores.

Systems-on-chip

- (c) Why does dark silicon lead system-on-chip designers to incorporate accelerators? Give an example of an accelerator to illustrate your answer. [4 marks]

Answer: Since we have so many transistors available, it makes sense to have some dedicated accelerators that can provide significant improvements in performance or power savings. A good example of this is a video codec like H.264 for phones and tablets. When playing a video, much of the SoC can operate in a low-power mode leaving the very efficient video codec to do all of the heavy computational work. This is far more efficient than getting a CPU or GPU to run the video codec algorithm.

Systems-on-chip

- (d) Why might systems-on-chip contain many more processor cores than the application class cores, which are typically the ones advertised? What characteristics do these cores possess? [4 marks]

Answer: Many I/O devices contain small microcontrollers. This provides flexibility including the ability to ship firmware updates to fix bugs. These processor cores are usually small and simple. There are also domain optimised cores for accelerators including GPUs, which are far more complex and can cover a large silicon area. Small low-power cores are also used for always-on functionality, real-time requirements and secure processors.

DRAM

- (e) Why do you need to perform both a row access and column read when reading data out of DRAM? What does each operation do? [3 marks]

Answer: A row access simply selects a whole row of bits from each array within a bank and places them in the sense amplifiers. (This assumes that a precharge operations has preceded

the row access.) The column read actually selects the correct bit or bits from each row, reading out as many bits per bank as there are arrays.

Memory hierarchy and caching (f) DRAM can perform burst reads and writes of several words. How does the last-level cache use and benefit from DRAM burst accesses? [3 marks]

Answer: Caches exploit spatial locality, a statistical observation that if a word is accessed then its neighbors are likely to be accessed soon. This leads us to having cache lines that contain several words of memory, and ties in nicely with DRAM burst reads to fill cache lines. DRAM burst writes are also convenient for spilling cache lines back to DRAM.
